CLINOMA PLATFORM

# FIRST PAPER CAMP

## Questions & Exercises Booklet

Day 3: Neurology & Cardiology (CVS)

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# CHAPTER 01

## Pediatric Neurology

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## 1. Short Essay / Enumerate

Q1: Enumerate four main characteristic weakness features of Duchenne Muscular Dystrophy (DMD).

Q2: Enumerate four prenatal factors involved in the etiology of Cerebral Palsy (CP).

Q3: Enumerate four diagnostic investigations used for Duchenne Muscular Dystrophy (DMD).

## 2. Short Answer Questions (Mention / List)

Q1: Mention four clinical signs or evaluation maneuvers used to assess a child with Floppy Infant Syndrome.

Q2: List four associated deficits that may accompany the core motor deficit in Cerebral Palsy (CP).

## 3. Extended Matching Questions (EMG)

**Options: A. Spastic CP B. Dyskinetic CP C. Ataxic CP D. Atonic (Hypotonic) CP E. Becker Muscle Dystrophy F. Duchenne Muscular Dystrophy G. Werdnig-Hoffman disease**

Match the following clinical descriptions to the correct option above:

1. A 4-year-old boy presenting with progressive proximal muscle weakness, calf pseudohypertrophy, markedly elevated creatine kinase (CK), and a history of a slowly progressive course with an X-linked recessive inheritance.

2. An infant exhibiting disturbances in balance and equilibrium, walking unsteadily with poor coordination, accompanied by hypotonia and nystagmus.

3. A child presenting with hypertonia, a positive Babinski sign (extensor plantar response), and persistent primitive reflexes.

4. An anterior horn cell disease that serves as a spinal cord lesion cause of Floppy Infant Syndrome.

## 4. Problem Solving (Cases)

## Case 1

A 3-year-old male child is brought to the clinic by his parents because he has difficulty climbing stairs and falls frequently. On examination, the clinician notes a waddling lordotic gait. When asked to rise from the floor, the child uses his hands to walk up his own legs to attain an erect posture. Prominent enlargement of the calf muscles is noted bilaterally, which feels firm rather than strong.

Q1: What is the most likely diagnosis?

Q2: What is the sign described when the child walks up his legs to stand?

Q3: Name the underlying protein defect responsible for this condition.

Q4: What diagnostic test serves as the definitive tool to confirm the muscle pathology?

**Case 2**

An 18-month-old child presents with a non-progressive central motor deficit. The mother reports a history of severe birth asphyxia during delivery. Physical examination reveals dynamic, involuntary, slow twisting movements along the long axis of the upper limbs, which become more prominent distally. Muscle tone fluctuates between increased and decreased states.

Q1: What is the general diagnosis for this child?

Q2: Which specific physiological classification of CP does this case represent?

Q3: Based on the timing of insult, which category does birth asphyxia fall under?

Q4: Is this neurological condition progressive or non-progressive?

## 5. Definitions

Q1: Define Duchenne Muscular Dystrophy (DMD).

Q2: Define Cerebral Palsy (CP).

Q3: Define Floppy Infant Syndrome.

## 1. Short Essay / Enumerate

Q1: Enumerate four physiological classifications (Major Motor Abnormalities) of Cerebral Palsy (CP).

Q2: Enumerate four anatomical and etiological causes of Floppy Infant Syndrome under the category of "Cerebral Causes".

Q3: Enumerate four criteria used to differentiate Central Hypotonia from Peripheral causes in a floppy infant.

## 2. Short Answer Questions (Mention / List)

Q1: Mention three associated features (comorbidities) that are constantly or frequently found with Duchenne Muscular Dystrophy (DMD).

Q2: List the treatment lines available for a patient with Duchenne Muscular Dystrophy (DMD).

## 3. Extended Matching Questions (EMG)

**Options: A. Hemiplegia B. Diplegia C. Quadriplegia / Tetraplegia D. Monoplegia E. Paraplegia F. Rigidity G. Chorea**

Match the following anatomical or physiological descriptions of Cerebral Palsy to the correct option above:

1. Arms and legs are equally involved, or arms are more involved than legs.

2. Arm and leg on the same side of the body are involved (Right or Left).

3. Quick irregular dysrhythmic movements involving proximal skeletal muscles.

4. Involvement of the legs only.

## 4. Problem Solving (Cases)

 

## Case 1

A 2-month-old infant is brought to the pediatric clinic because the mother feels the baby is "too soft and limp" compared to her older siblings. On examination, when the baby is placed in a supine position, his lower limbs rest flat on the bed in a "Frog Leg Position". When pulling the infant up from his hands into a sitting position, the head completely falls backward. Nerve Conduction Velocity (NCV) and Electromyography (EMG) are ordered to differentiate lower motor unit disorders from central conditions.

Q1: What is the provisional diagnosis for this clinical presentation?

Q2: What does the "Head Lag" sign denote during evaluation?

Q3: If the NCV and EMG confirm a lower motor unit disorder with a neuropathy pattern, what characteristic would you expect to find regarding deep tendon reflexes?

Q4: Mention one anterior horn cell disease that could cause this presentation.

**Case 2**

A 5-year-old child presents with a non-progressive central motor deficit. Examination shows dynamic deformities and a history pointing toward a perinatal birth trauma. A multidisciplinary team is assigned to manage the patient. The pediatrician prescribes muscle relaxants and sedatives to manage the child's severe spasticity.

Q1: What is the core diagnosis?

Q2: Why are muscle relaxants and sedatives prescribed by the pediatrician in this case?

Q3: Name two other specialists who should be part of this multidisciplinary management team.

Q4: Why might this child be referred to an orthopedic surgeon?

## 5. Definitions

Q1: Define Lower Motor Unit Disorders Criteria in the context of Floppy Infant Syndrome.

Q2: Define Becker Muscle Dystrophy (BMD).

## CHAPTER 02

# Pediatric Cardiology (CVS)

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## 1. Short Essay / Enumerate

Q1: Enumerate the 4 Classic Components of Tetralogy of Fallot (TOF).

Q2: Enumerate four absolute contraindications or side effects/toxicity signs of Digoxin therapy (Digitalis Protocol) in Pediatric Heart Failure.

Q3: Enumerate four definitive closure indications and timing criteria for a patient with a Ventricular Septal Defect (VSD).

## 2. Short Answer Questions (Mention / List)

Q1: Mention the components of "The Essential Triad" in the clinical picture of Heart Failure in infants.

Q2: List four early closure indications (before 2 years of age) for a patient with an Atrial Septal Defect (ASD).

## 3. Extended Matching Questions (EMG)

 

**Options:**

1. A. Ventricular Septal Defect (VSD)
2. C. Patent Ductus Arteriosus (PDA)
3. E. Tetralogy of Fallot (TOF)
4. Match the following clinical descriptions or diagnostic findings to the correct option above:
5. Answer: D (Complete Transposition of the Great Arteries)
6. Answer: C (Patent Ductus Arteriosus)
7. Answer: E (Tetralogy of Fallot)
8. Answer: A (Ventricular Septal Defect)

## Case 1

A 7-month-old infant is brought to the emergency department during a crying episode. The mother notes that the child suddenly became deeply cyanotic, agitated, and started breathing very rapidly. On physical examination, the child is tachypneic and irritable. Auscultation reveals a systolic ejection murmur at the left sternal border that softens significantly during this acute episode. The emergency team immediately places the infant in a Knee-Chest position and prepares to administer supplemental oxygen and Morphine Sulfate.

Q1: What is the diagnosis for this acute clinical episode?

Q2: What is the underlying congenital heart disease?

Q3: Why does the systolic ejection murmur soften or disappear during this spell?

Q4: Mention two medications or interventions that are strictly contraindicated during this event.

### Case 2

An 8-week-old infant with a known history of a large Ventricular Septal Defect (VSD) presents with delayed growth, excessive sweating during feeding, and recurrent chest infections. On examination, the infant has tachypnea and a heart rate of 175 bpm. The liver is palpable 4 cm below the costal margin and is tender. The pediatrician diagnoses Congestive Heart Failure (CHF).

Q1: According to the Ross Classification for heart failure, what class does this infant belong to?

Q2: What is the drug of choice recommended for preload reduction in this patient?

Q3: What type of nutritional management is indicated to support growth?

Q4: What class of afterload-reducing agents is vital for infants with large left-to-right shunts?

## 5. Definitions

Q1: Define Eisenmenger Syndrome (in the context of left-to-right shunts).

Q2: Define Pediatric Heart Failure (HF).

Q3: Define Ostium Secundum Defect.

## 1. Short Essay / Enumerate

Q1: Enumerate the four Core Mechanisms & Causes of Pediatric Heart Failure (HF) based on its pathogenesis.

Q2: Enumerate four clinical manifestations found under the category of "Systemic Congestion (Right-Sided Heart Failure)" in older children.

Q3: Enumerate four features or symptoms observed during the "Eisenmenger's Phase" of a moderate to large Ventricular Septal Defect (VSD).

## 2. Short Answer Questions (Mention / List)

Q1: Mention four precipitating factors that can lead to Pediatric Heart Failure (HF).

Q2: List three physiological and anatomical types of Atrial Septal Defect (ASD).

## 3. Extended Matching Questions (EMG)

**Options:**

1. A. Membranous VSD
2. C. Infundibular VSD
3. E. Ostium Secundum Defect
4. Match the following anatomical or auscultatory descriptions to the correct option above:
5. Answer: B (Muscular VSD)
6. Answer: E (Ostium Secundum Defect)
7. Answer: C (Infundibular VSD)
8. Answer: A (Membranous VSD)

### Case 1

A 3-week-old male infant, born full-term, presents to the neonatal intensive care unit with acute, rapid onset of congestive heart failure symptoms, progressive dyspnea, and severe feeding difficulties. On clinical examination, the patient exhibits intense cyanosis that was noted within the first 2 days of life and fails to show any improvement when placed under $100\%$ oxygen therapy. Systemic congestion with noticeable hepatomegaly is noted. Echocardiography is immediately performed as the primary diagnostic modality.

Q1: What is the most likely diagnosis for this infant? * Complete Transposition of the Great Arteries (TGA).

Q2: Why does this condition carry a survival obligation incompatible with life without mixing sites? * Because it creates two independent circulations running in parallel rather than in series.

### Case 2

A 4-month-old female infant is evaluated for poor weight gain and delayed growth. Her mother reports that the baby sweats excessively during feedings and breathes very fast. Cardiovascular examination reveals a hyperkinetic apex, a precordial bulge, and a palpable systolic thrill at the lower left sternal border. Auscultation confirms a harsh pansystolic murmur at the lower left sternal border. A chest X-ray reveals cardiomegaly with clear enlargement of the left atrium and left ventricle along with increased pulmonary vascular markings.

Q1: What congenital cardiac defect is most consistent with this presentation? * Moderate to Large Ventricular Septal Defect (VSD).

Q2: Which chamber enlargement is typically identified first on an ECG in a moderate VSD? * Left Ventricular Hypertrophy (LVH) & Left Atrial Hypertrophy (LAH).

## 5. Definitions

Q1: Define Patent Ductus Arteriosus (PDA).

Q2: Define Hypercyanotic (TET) Spells.

Q3: Define Ross Classification Class IV.